

Simulation of Hydrogen Degassing using Vacuum Arc Degasser

Date : 05-02-2025

Simulation of Hydrogen degassing using a Vacuum arc degasser with an argon injection plug. This simulation is performed using Ansys Fluent supplemented by UDFs.

Domain : Ladle Dimensions

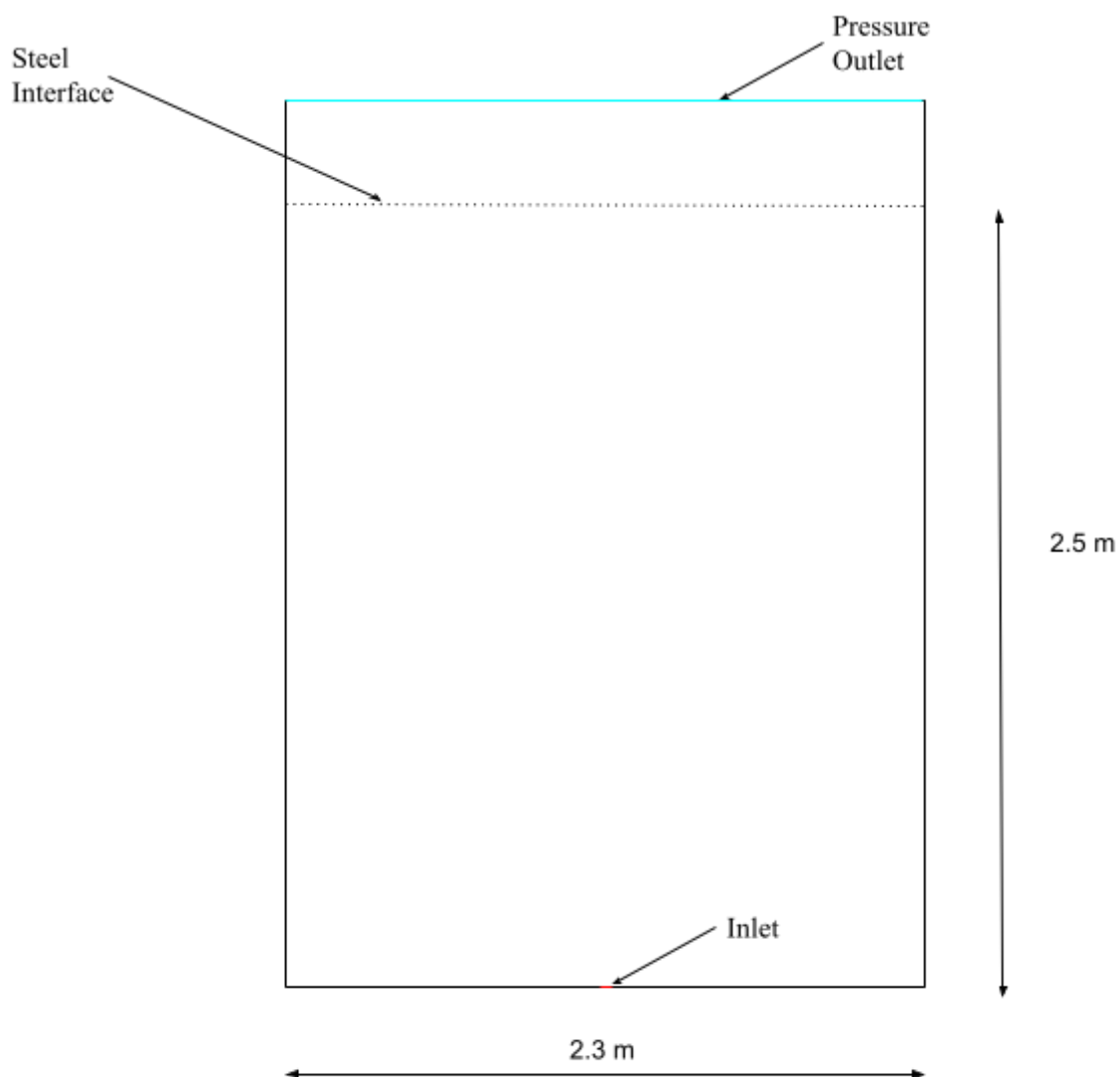


Figure 1: Ladle dimensions and domain setup

The Inlet for argon injection is currently centred. Usually, a porous plug is used for the inlet. In this simulation, we are using an open inlet, with a diameter of 18 mm. Assuming a porous inlet of 60 mm was used, this inlet would be equivalent to the same with 9% openings for flow to pass through.

Setup:

1. Mesh

A poly-hexcore mesh is created using fluents meshing. The mesh contains 190,106 cells. The initial mesh contained only 50,000 but was improved via inbuilt mesh adaptation in fluent after initial flow stabilization.

2. Multiphase Model

The Eulerian Multiphase model is utilized without modifying default settings on the model selection. The population balance method is enabled. In this simulation, considering a flow rate of 600ml/s, the droplet size is set between 0.01 m and 0.08 m. The expected droplets should be in between 0.06 to 0.03 based on the reference article. Initially, only the Schiller naumann drag model is enabled. Upon stabilization of flow, approximately after 15 seconds of simulation time the life model (tomiyama) and the turbulent dispersion model can be enabled.

3. Turbulence Model

The standard $\kappa - \epsilon$ Model is utilized with standard wall treatment. Production limiter is enabled initially. It is advisable to run using the dispersed mode first, and then shift to the per phase mode, once the has stabilized.

4. Species Transport

Species transport is enabled. The mixture phases are set in the materials tab. The liquid-steel phase will contain liquid steel and hydrogen, argon phase will contain hydrogen and argon as species. It is important to ensure that hydrogen is set as the first species. The slag phase can be used as pure fluid as no species transport is considered from slag. The following table can be referred for material properties.

Parameter	Value
Slag thickness (m)	0.1
Slag viscosity ($\text{kgm}^{-1} \text{s}^{-1}$)	0.04
Slag density (kgm^{-3})	3000
Steel viscosity ($\text{kgm}^{-1} \text{s}^{-1}$)	0.005
Steel density (kgm^{-3})	7000
Argon viscosity ($\text{kgm}^{-1} \text{s}^{-1}$)	2.12e-5
Argon density (kgm^{-3})	1.62
Argon/Steel surface tension (Nm^{-1})	1.19
Slag/Argon surface tension (Nm^{-1})	0.58
Slag/Steel surface tension (Nm^{-1})	0.12
Temperature (K)	1871

Table 1 : The fluid properties are set using the above data from the reference article.

5. Boundary Conditions

Inlet: - velocity is 2.36 m/s . Argon volume fraction is 1, inlet droplet diameter is around 0.04 m. The species' hydrogen mass fraction is zero.

Outlet : - Pressure outlet set at 0 gauge pressure. Back flow Volume fraction of argon-h2 is 1. The Backflow species is argon.

Walls : - no slip - liquid steel

Slip (zero shear) - argon

Operating conditions : - Operating pressure is set as 1013 Pa (1% of atm) at a reference location above the interface. Gravity -> -9.81 y, and the density calculation is set as minimum phase averaged.

6. User Defined Functions

Two User defined functions are currently utilized.

1. Density :- Sets up the density of argon-hydrogen mixture based on hydrostatic location.

2. Mass Transfer :- using the DEFINE_MASS_TRANSFER macro, which is for species transport between multiphase flows, we implement the sieverts law, and calculate the Source for the transport equation.

This has to be setup in the multiphase interactions panel, and non-linear udf has to be enabled from TUI using /solver/set/expert.

7. Methods and Controls

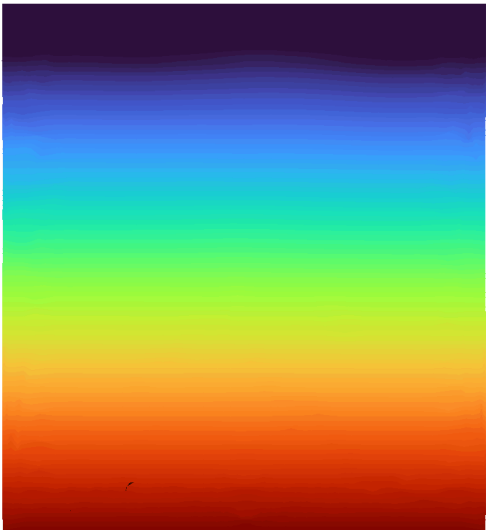
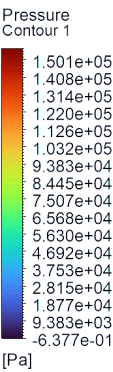
Use the *Coupled Solver*, with solve N-Phase volume fraction equations enabled. For Gradient, the least squares cell based scheme is used. For pressure, the Body Force weighted discretization scheme has been found to be stable. For all other equations, the first order upwind schemes is utilized, at least for the initial flow stabilization period. Enable High order term relaxation and Warped face gradient correction.

In terms of controls, URFs under 0.5 are used, and the flow courant number is set to 25.

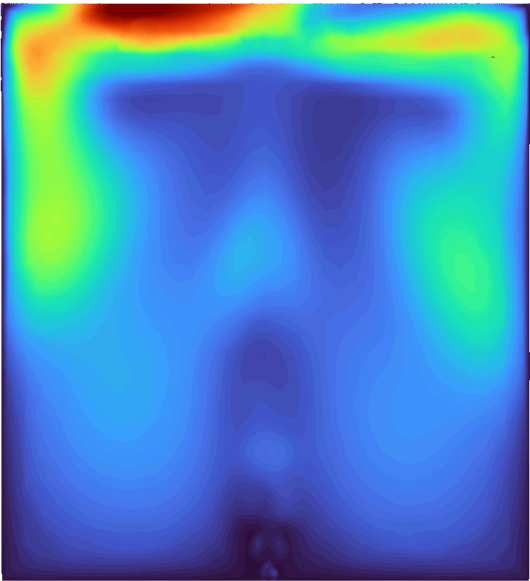
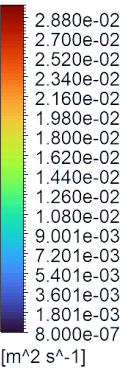
8. Initialization

Initialize with 0 Gauge pressure, 0 velocity. Ensure that hydrogen mass fraction is 5e-06 (5 ppm), and argon has 0 mass fraction of hydrogen. Then, patch the above interface (Gas-vacuum region) to have 1 Volume fraction of argon-h2 mixture. Also set the droplet size to maximum bin size, as the droplet is expected to expand upon entering the low pressure vacuum region.

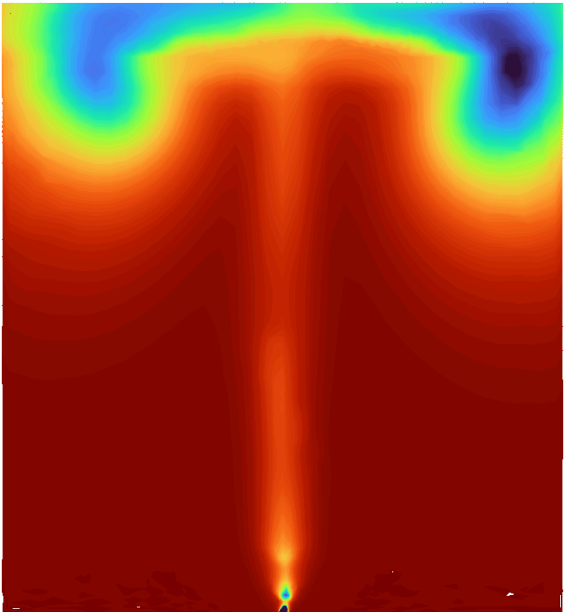
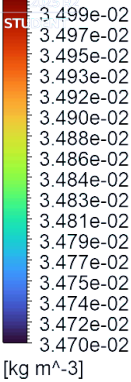
Simulation Results : 11 Seconds :-

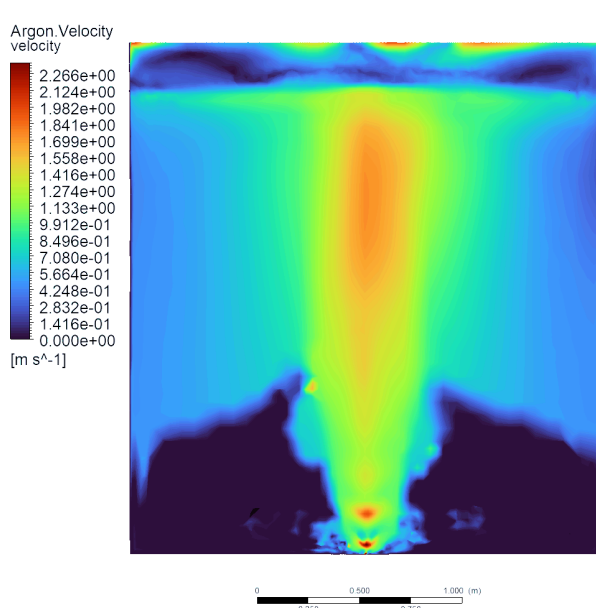


Liquid Steel.H2.Effective Diffusion Coefficient
h2 content

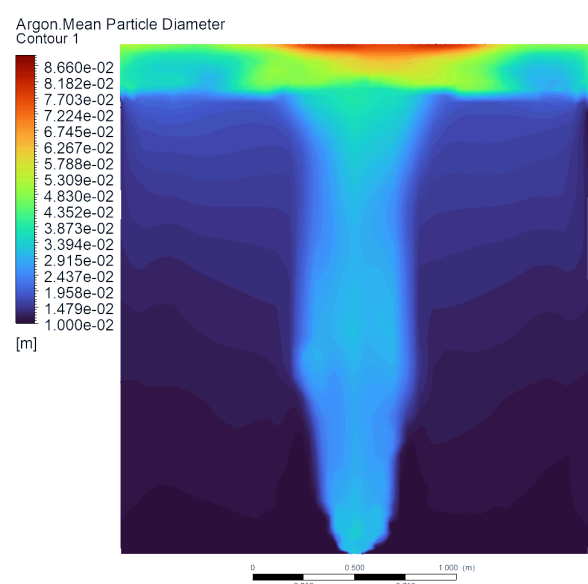


Liquid Steel.Mass Concentration of H2 Phase Level
H2 mass fraction in liquid steel





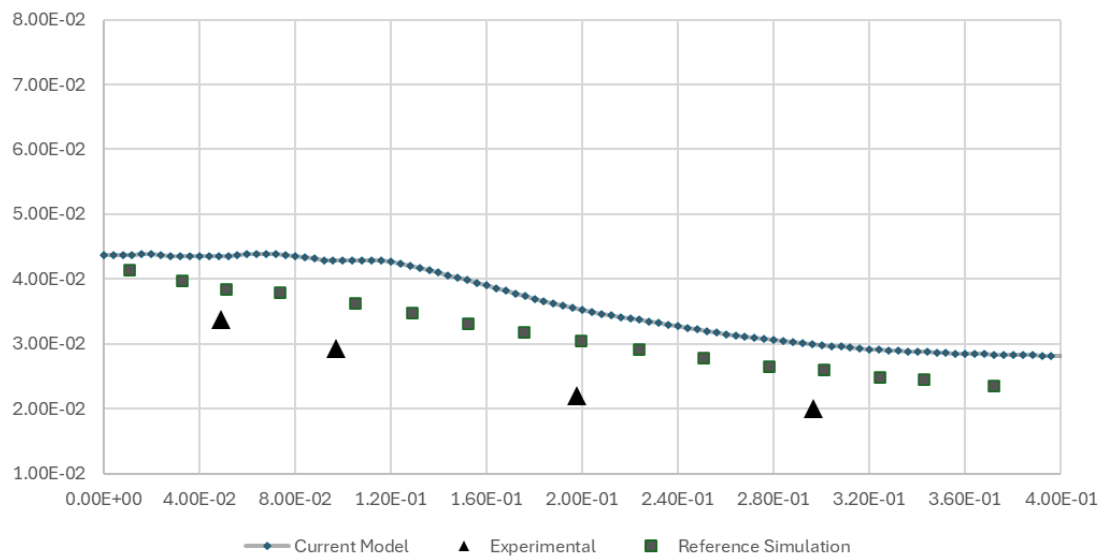
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Validation

Comparison of model with reference (650K cells)



This plot compares the results of our models results to the simulation model in the reference article. The experimental data also comes from the paper. Testing with increased mesh resolution is ongoing to improve the validation.